

Operations Research

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Solve the given linear programming problem (LPP) by using simplex method and interpret the findings. Maximize $Z = 4A + 3B + 6C$ Subject to the constraints: $2A + 3B + 2C \leq 44$, $4A + 3C \leq 47$, $2A + 5B \leq 43$, and $A, B, C \geq 0$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. A project consists of nine activities whose time estimates (in weeks) and other characteristics are given below:

Activities	A	B	C	D	E	F	G	H	I
Preceding activities	---	A	A	B	B	D	D	F	F
Optimistic time	26	62	11	83	94	46	12	51	41
Most likely time	46	12	51	41	106	15	10	66	24
Pessimistic time	66	24	82	31	29	27	16	6	16

(a) Show the PERT network diagram for the project. (b) Identify the critical activities. (c) What is the expected project completing time and its variance?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. A milk salesman estimates the probability of the demand for a litre of milk is as follows:

Demand	1	11	21	31	41	5
Probability	0.1	0.0	0.15	0.3	0.25	0.20

He purchases a litre of milk @ of Rs. 60 and sells it @ of Rs. 70. Assuming that unsold milk has no scrap value, find (a) Find optimum quantity that would obtain Max. EMV. (b) Find the minimum value of EOL. (c) What is the value of expected profit with perfect information (EPPI)?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. The ABC company has three jobs to be done on three machines. Each job must be done on one and only one machine. The cost of each job on each machine is given in the following table.

Machine	Jobs	X	Y	Z
A	4	6	8	2
B	2	3	4	8
C	4	8	5	1

By using Hungarian method, find the job assignments which will minimize the cost.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Food X contains 6 units of vitamin A and 7 units of vitamin B and costs Rs. 5 per gram. Food Y contains 8 units of vitamin A and 12 units of vitamin B and costs Rs. 18 per gram. The daily minimum requirement of vitamins A and B are respectively 100 units and 138 units respectively. Formulate the problem as a LPP with the objective function minimizing the cost.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Customers arrive at a bank having single counter at the rate of 25 customers per hour. Time required to serve a customer has exponential distribution and average number of customer served per hour is 30. Find the average number of customers in queue and in system as well.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find the initial solution by using Vogel's Approximation Method (VAM). From To PQR Supply A 193050107 B 703040609 C 408702018 Demand 5871434

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. The following is the payoff matrix of a game being played by A and B. Determine the optimal strategies for the players and the value of the game. A's strategies B's strategies B ■ B ■ A ■ 120- 80- 20A ■ 607030A ■ -1007020

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Draw the network diagram and identify critical path and bottleneck activities. Activity A B C D E F G Predecessors - A A B, C C, D E, F Time (in days) 7 4 5 6 7 5 6

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Describe dominance rule method for solving a problem of game theory.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Describe different types of queuing disciplines used in serving a customer in a queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on: (a) Marginal analysis approach in decision making. (b) Modified distribution (MODI) method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.